



**MACAU UNIVERSITY OF SCIENCE AND  
TECHNOLOGY**

**School of Computer Science and Engineering  
Faculty of Innovation Engineering**

**<<Software Project for Course Software Engineering>>**

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# **1. Introduction**

## **1.1. Purpose**

This document is a detailed system requirements specification for a defined service.

This document will build on the Business Requirements Specification (BRS) and must have a corresponding High-Level Design (HLD) document. Any required changes to the Business Requirements Specification or the High-Level Design that arise in developing this System Requirements Specification (SRS) must be updated in those documents.

Current and valid System Requirements Specification (SRS), High-Level Design (HLD), and Low-Level Design (LLD) are required as part of the operational support documentation set for any application.

## **1.2. Notation**

This specification and its supporting documents and diagrams use the Unified Modelling Language notation.

## **1.3. Scope**

This section describes the scope of the system.

## **1.4. Context Diagram**

A context diagram is included to demonstrate how the scope of this system integrates with related systems.

## **1.5. Definitions and Acronyms**

This section describes any terms used.

## **1.6. References**

This section lists any references and includes pre-requisite documentation and any other supporting documentation such as diagrams.

## **1.7. Overview**

This section describes the system requirements and explains how the references listed above can help support that overview.

# **2. General Description**

## **2.1. System Functions**

This section describes the high level purpose and functions of the system.

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## 2.2. User Characteristics

This section describes the various user types (roles) and their characteristics. The type of user interface that is required (data entry, self-service, trained user, global, organisation). This information will inform the Low Level Design. This includes information on predicted user platforms (devices, operating systems and browsers).

## 3. General Constraints

### 3.1. Software Constraints

This section lists the software constraints. This may cover licensing, technical standards, preferred products, and the impact of given decisions.

### 3.2. Hardware Constraints

This section lists the hardware constraints. This may cover licensing, technical standards, preferred products, and the impact of given decisions.

## 4. Assumptions and Dependencies

This section describes the assumptions and dependencies made in forming this system requirement.

## 5. Functional Requirements Master List

This section lists the functional requirements in summary form. Each of these will be further defined in subsequent sub-sections.

| Req. ID | Requirement Name | Requirement Description |
|---------|------------------|-------------------------|
| REQ-1.1 |                  |                         |
|         |                  |                         |
|         |                  |                         |
|         |                  |                         |
|         |                  |                         |
|         |                  |                         |
|         |                  |                         |

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## **6. Functional Requirement REQ-1.1**

### **6.1. Description**

A description of this requirement

### **6.2. System Input**

Describe the inputs to this requirement.

### **6.3. Display**

Describe the display requirements in terms of what is to be achieved by the operator. Include filtering, grouping, ordering, and specific data items to display. The display may include reporting of some form (screen, PDF, Excel) or reference to the reporting solution.

### **6.4. System Processing**

Include the system processing including any business rules.

### **6.5. System Output**

Describe the outputs from this requirement.

### **6.6. Other**

### **6.7. Constraints**

State any functional-specific constraints

### **6.8. Data Handling**

Describe any data handling such as validations, calculations, and data formats such as the use of ASCII or Unicode.

### **6.9. Error Handling**

Reference any common error handling. Detail how known or special errors will be handled especially where that differs from the common error handling process.

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## 7. External Interface Requirement

### 7.1. Data Interfaces

### 7.2. User Interfaces

### 7.3. Other Interfaces

Describe any hardware, software, communications, or other interfaces not covered elsewhere.

## 8. Non-Functional Requirements

### 8.1. System Performance

This section describes the system performance requirements. Example response times are suggested though different systems requirements may apply.

| Req. Id  | Description                                                                                                             | Response Time         |
|----------|-------------------------------------------------------------------------------------------------------------------------|-----------------------|
| PERF-1.1 | Time experienced by the operator for regular tasks such as moving from one field to another                             | 1 second              |
| PERF-1.2 | Time experienced by the operator for intermediate system processing such as opening a new screen or submitting data.    | 3 seconds             |
| PERF-1.3 | Time experienced by the operator for significant system processing such as credit card authorization                    | 7 seconds             |
| PERF-1.4 | Time experienced by the operator for a process response where integration with other systems and processes is required. | 2 hours?<br>24 hours? |
|          |                                                                                                                         |                       |

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## 8.2. Information Security

This section describes the information security requirements.

| Req. Id | Description                                                                                                                                                                                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-1.1 | User credentials and data covered by the Data Protection Act must not be transmitted in plain text over any network.                                                                                                                            |
| SEC-1.2 | Data covered by the Data Protection Act shall not be held in environments that have not passed full system testing.<br><br>This means any environments other than PRODUCTION or PRE-PRODUCTION as described by UCL's Release Management Policy. |
| SEC-1.3 | Shall meet requirements of UCL's Information Security policies.                                                                                                                                                                                 |
| SEC-1.4 |                                                                                                                                                                                                                                                 |

## 8.3. Availability

This section describes the availability requirements in terms of days and permissible planned and unplanned unavailability.

| Req. Id | Description |
|---------|-------------|
| AVA-1.1 |             |
| AVA-1.2 |             |
| AVA-1.3 |             |
| AVA-1.4 |             |



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## 8.4. Capacity

| Req. Id | Description                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| CAP-1.1 | The system must be able to manage the following data volumes in year ONE. <ul style="list-style-type: none"><li>•</li></ul>                        |
| CAP-1.2 | The system must be able to manage the following data volumes in year TWO. <ul style="list-style-type: none"><li>•</li></ul>                        |
| CAP-1.3 | The system must be able to manage the following data volumes in year THREE. <ul style="list-style-type: none"><li>•</li></ul>                      |
| CAP-2.1 | The system must be able to manage the following concurrent user operation volumes in year ONE. <ul style="list-style-type: none"><li>•</li></ul>   |
| CAP-2.2 | The system must be able to manage the following concurrent user operation volumes in year TWO. <ul style="list-style-type: none"><li>•</li></ul>   |
| CAP-2.3 | The system must be able to manage the following concurrent user operation volumes in year THREE. <ul style="list-style-type: none"><li>•</li></ul> |
| CAP-3.1 |                                                                                                                                                    |